

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I C SANGITA(192373059) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Csangita

Annexure A

1. Secure Password Storage in Web Applications

A web application stores user passwords and needs to protect them against database breaches.

- A. Explain why storing plaintext passwords is insecure.
 - B. Suggest a secure method for storing passwords and justify your choice.
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2. Digital Signature for Document Authentication

A company wants to ensure that official documents sent electronically are authentic and unaltered.

- A. Explain how digital signatures can be used in this scenario.
 - B. Justify how this method ensures both integrity and non-repudiation.
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3. Preventing Replay Attacks in Online Transactions

An online payment system is vulnerable to replay attacks where attackers resend previously captured transactions.

- A. Explain how replay attacks occur.
 - B. Suggest a cryptographic technique to prevent such attacks and justify your answer.
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4. Secure Communication over Public Wi-Fi

A user accesses sensitive information using a public Wi-Fi network without encryption.

- A. Identify the risks involved in this scenario.
 - B. Recommend a secure communication method and justify how it protects user data.
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5. Data Integrity in Cloud Storage

An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

- A. Explain how hashing can be used to verify data integrity.
- B. Suggest an additional mechanism to enhance trust and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS:

2.Digital Signature for Document Authentication

A company wants to ensure that official documents sent electronically are authentic and unaltered.

A. Explain how digital signatures can be used in this scenario.

B. Justify how this method ensures both integrity and non-repudiation.

Solution:

Digital signature for document authentication

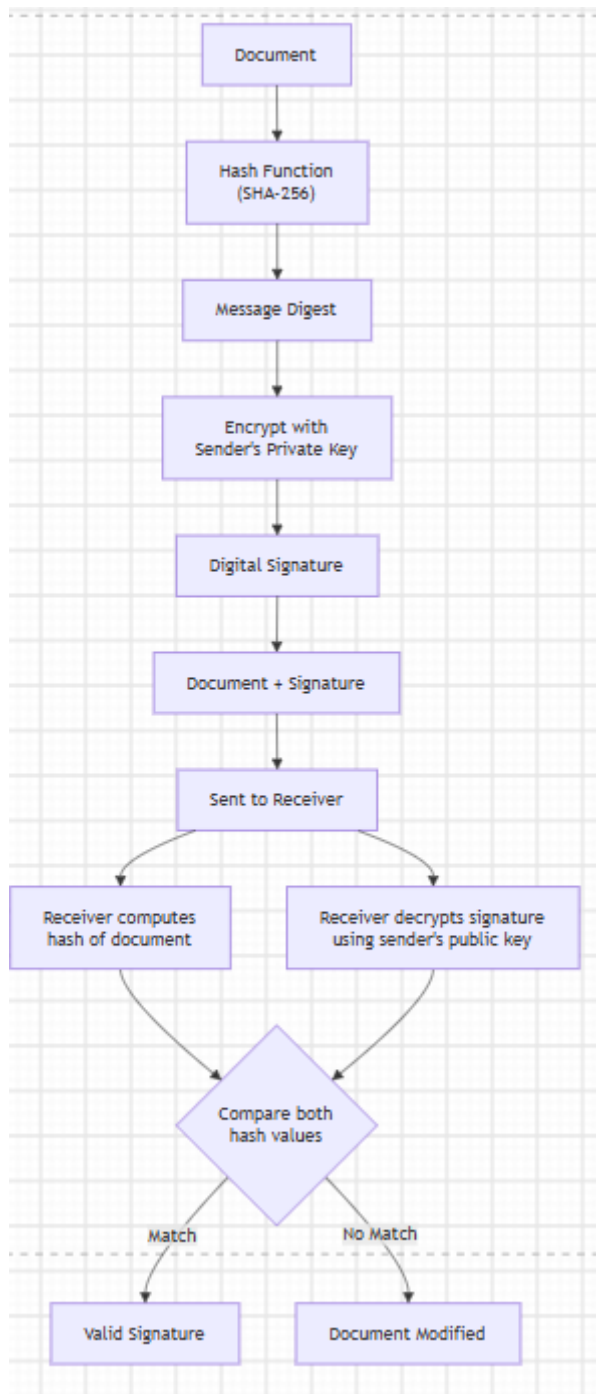
Introduction

A digital signature is a cryptographic technique used to verify the authenticity of an electronic document, message, or file. It is based on public key cryptography, where a user has a private key and a public key. The private key is used to create the digital signature, and the public key is used to verify it.

In this scenario, the company wants to ensure that official documents sent electronically are authentic and have not been altered during transmission. Digital signatures provide authentication, integrity, and non-repudiation.

Digital signature process

The sender signs the document using their private key, and the receiver verifies it using the sender's public key.



Step-by-step working

Step 1: Create the document

The sender prepares an electronic document, such as a contract, invoice, or company notice.

Step 2: Generate a hash

A hash function such as SHA-256 is applied to the document.

Example:

Document: "Project approved"

SHA-256 hash:

A1B2C3D4E5F6...

The hash is called the message digest.

Step 3: Create the digital signature

The sender encrypts the message digest using their private key.

Signature = Encrypt(Hash, Private Key)

This encrypted hash becomes the digital signature.

Step 4: Send the document

The sender sends:

- Original document
- Digital signature

Step 5: Verify the signature

The receiver performs two operations:

1. Computes the hash of the received document.
2. Decrypts the digital signature using the sender's public key.

If both hash values are equal, the document is authentic.

Example using RSA

Suppose:

- Private key = (d, n)
- Public key = (e, n)

Hash value:

$$H = 1234$$

The sender signs:

$$S = H^d \bmod n$$

The receiver verifies:

$$H = S^e \bmod n$$

If the recovered hash matches the newly calculated hash, the signature is valid.

How digital signatures ensure integrity

Integrity means that the document has not been modified.

When the receiver calculates the hash again, even a small change in the document produces a completely different hash value.

Example:

Original:

Amount = Rs. 50,000

Modified:

Amount = Rs. 80,000

The hash changes completely.

Therefore:

- Hash mismatch
- Signature verification fails
- Receiver detects tampering

How digital signatures ensure non-repudiation

Non-repudiation means that the sender cannot deny sending the document.

This is achieved because:

- Only the sender possesses the private key.
- The signature was created using that private key.
- Anyone can verify it using the public key.

Therefore, the sender cannot later claim:

"I did not send this document."

This provides legal and forensic evidence.

Authentication using digital signatures

Authentication confirms the identity of the sender.

The receiver verifies the signature using the sender's public key.

A successful verification proves that:

- The sender owns the corresponding private key.
- The document originated from the claimed sender.

Security analysis

Advantages

- Verifies sender identity
- Detects document modification
- Prevents forgery
- Supports legal electronic documents
- Uses strong public key cryptography
- Can be verified by anyone with the public key

Limitations

- Private key compromise can create fake signatures
- Requires certificate management
- Computationally slower than symmetric encryption
- Depends on secure key storage

Applications

Digital signatures are widely used in:

- Government documents
- Banking transactions
- Online contracts
- Income tax filing
- E-commerce
- Software distribution
- Email security
- Cloud document signing

Comparison with handwritten signatures

Handwritten Signature	Digital Signature
Can be copied	Cryptographically protected
Difficult to verify	Easy to verify
No integrity protection	Detects modifications
Weak non-repudiation	Strong non-repudiation
Paper-based	Electronic

Conclusion

Digital signatures provide a secure method for authenticating electronic documents. The sender signs the document using a private key, and the receiver verifies it using the public key. Because the signature is based on the document hash, any modification changes the hash value and causes verification to fail. This guarantees integrity by detecting tampering and non-repudiation by preventing the sender from denying the signed document. Therefore, digital signatures are an essential component of modern cryptographic security systems and electronic document authentication.

5. Data integrity in cloud storage

An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

A.Explain how hashing can be used to verify data integrity.

B. Suggest an additional mechanism to enhance trust and justify your answer.

Introduction

Cloud storage allows organizations to store files and important information on remote servers. Although cloud storage provides easy access and scalability, organizations must ensure that stored data is not modified, corrupted, or tampered with. Data integrity means that the data remains accurate and unchanged from the time it was stored.

A common cryptographic technique used to verify data integrity is hashing. A hash function generates a fixed-size value called a hash value or message digest. By comparing hash values before and after storage, organizations can detect any unauthorized modification.

Hashing for data integrity

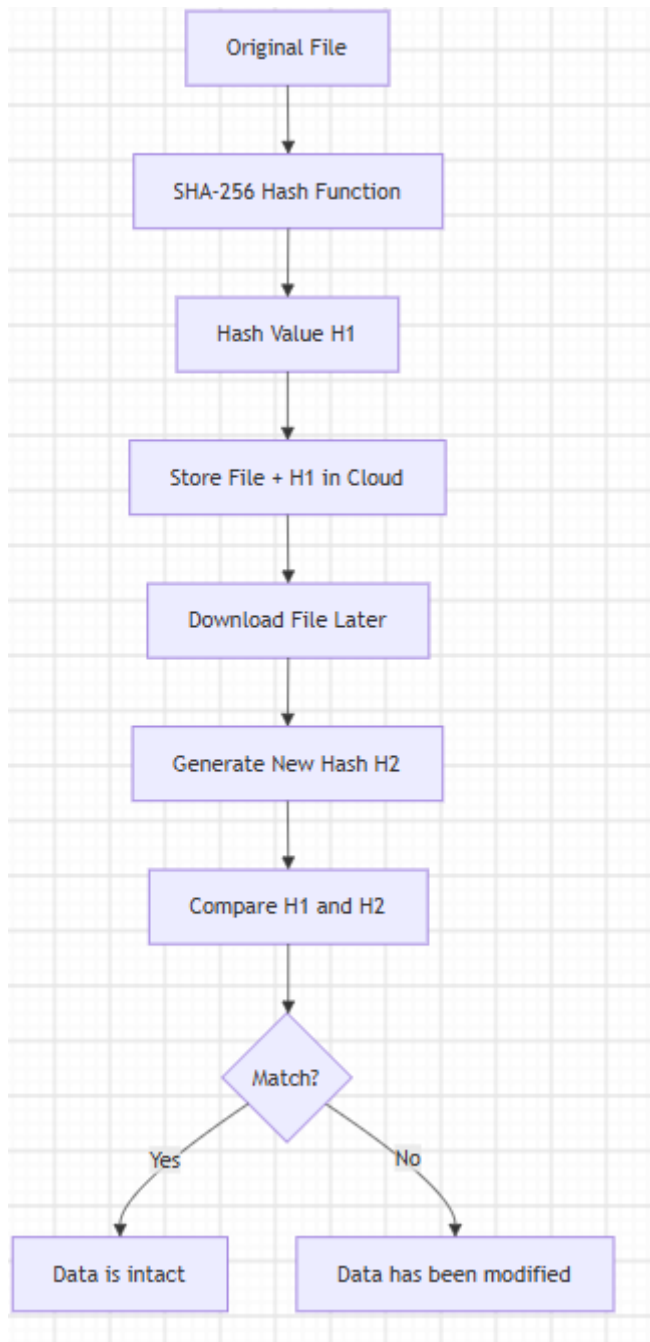
A hash function converts input data of any size into a fixed-length hash value.

Common hash algorithms:

- SHA-256
- SHA-3
- SHA-512

If the data changes even slightly, the hash value changes completely.

Integrity verification process



Step-by-step working

Step 1: Generate the original hash

Before uploading a file to the cloud, the organization computes its SHA-256 hash.

Example:

File:

Financial_Report.pdf

Hash:

A3F9C7E4B1D8...

This hash is stored securely.

Step 2: Upload the file

The original file is uploaded to the cloud server.

Step 3: Download the file

When the file is retrieved, the organization computes a new hash.

New hash:

A3F9C7E4B1D8...

Step 4: Compare hashes

If:

$H1 = H2$

the file is unchanged.

If:

$H1 \neq H2$

the file has been altered.

Example

Original file:

Balance = Rs. 10,00,000

Original hash:

8A91C3...

Modified file:

Balance = Rs. 90,00,000

New hash:

F71B9D...

Since the hashes are different, integrity verification fails.

Advantages of hashing

- Detects unauthorized modifications
- Fast computation
- Fixed-size output
- Efficient for large files
- Easy comparison of data
- Widely supported in cloud systems

Limitation of hashing

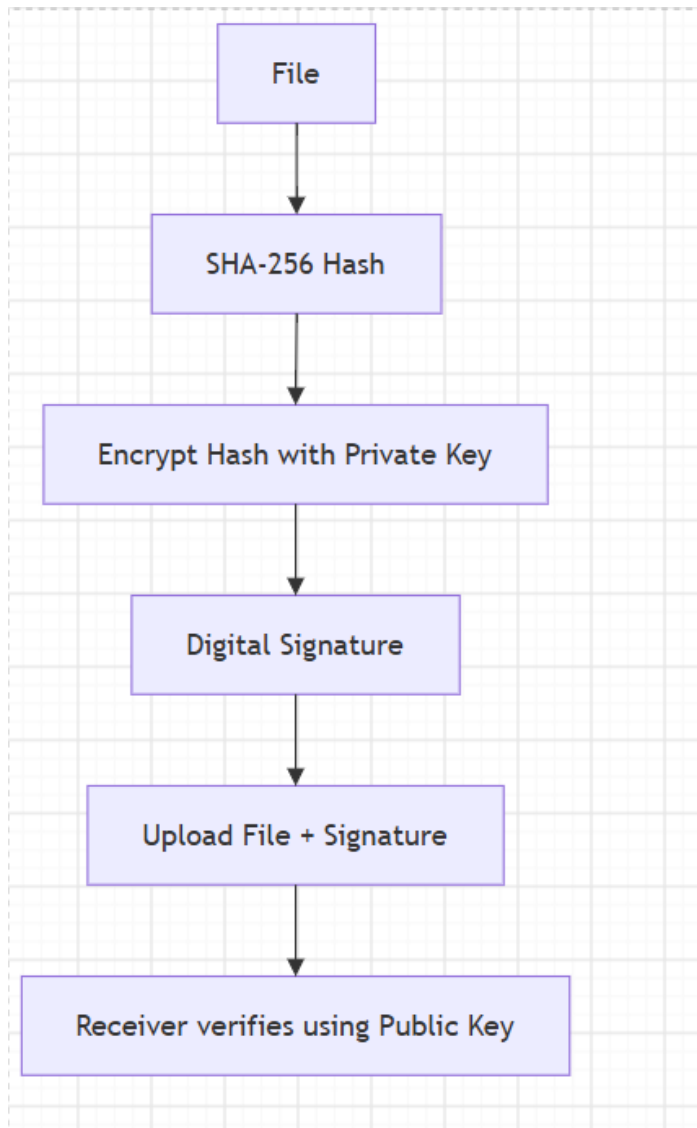
Hashing alone verifies integrity but does not prove who created the hash. An attacker who modifies both the file and the stored hash could bypass simple integrity checking.

Therefore, an additional security mechanism is required.

B. Additional mechanism: Digital signature

To enhance trust, the organization should use digital signatures together with hashing.

Digital signature process



Working

1. Compute the file hash.
2. Encrypt the hash using the owner's private key.
3. Store the file and the digital signature.
4. During verification:
 - Generate a new hash.
 - Decrypt the signature using the public key.
 - Compare both hashes.

Why digital signatures enhance trust

Digital signatures provide:

Integrity

Any modification changes the hash and causes verification failure.

Authentication

The signature proves the identity of the data owner.

Non-repudiation

The owner cannot deny signing the file.

Protection against hash replacement attacks

An attacker cannot generate a valid signature without the private key.

Comparison

Hashing	Digital Signature
Verifies integrity	Verifies integrity and authenticity
No authentication	Authentication provided
No non-repudiation	Non-repudiation provided

Security analysis

Using hashing alone:

- Detects accidental corruption
- Detects transmission errors
- Detects unauthorized modification

Using hashing with digital signatures:

- Detects modification
- Verifies the source
- Prevents forgery
- Increases trust in cloud storage

Real-world applications

- Cloud backup services
- Digital document storage
- Government records
- Medical records
- Financial databases
- Software distribution
- Blockchain systems

Conclusion

Hashing is an effective method for verifying data integrity in cloud storage because any change in the file produces a different hash value. By comparing the original and current hashes, organizations can detect unauthorized modifications. However, hashing alone cannot verify the identity of the data owner. Therefore, combining hashing with digital signatures provides a stronger security solution by ensuring integrity, authentication, and non-repudiation. This combination is the most reliable approach for protecting critical cloud-stored data.